U-45 CALCULATORS

User Manual

Calculator #26 Programmer Base-N Scientific (terminal green)

Scientific - natural display



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#26 — Programmer Base-N Scientific (terminal green)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 HEX]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

integer algebraic entry in the selected base (HEX/DEC/OCT/BIN), then =.

Chapters

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1 · Getting started

Open U-45 CALCULATORS, choose **#26** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#25, #26, #27, #31, #32, #33, #37, #38, #39, #43, #44, #45, #49, #50, #51, #53, #55
op	Operators & execute	#28, #29, #35, #40, #46, #52, #57, #58
fn	Functions	#1, #2, #3, #4, #5, #6, #8, #9, #10, #11, #13, #14, #15, #16, #17, #18, #19, #20, #21, #22, #23, #24, #34, #56
mem	Memory, variables & registers	#41, #42, #54
clr	Clear, delete & power	#12, #30, #36

mod	**Shift, modes, menus & navigation**	**#7, #47, #48**

2 · Features

47 features: 33 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE Exact integer arithmetic on BigInt values (no rounding at any word size)
2. LIVE HEX, DEC, OCT and BIN entry and display modes
3. LIVE Hex digits A–F on dedicated keys (refused in DEC/OCT/BIN)
4. LIVE Simultaneous view of the result in the three other bases
5. LIVE Word size 8 / 16 / 32 / 64 bits with wrap-around
6. LIVE Signed (two's complement) or unsigned interpretation
7. LIVE AND, OR, XOR bitwise operators
8. LIVE NAND, NOR, XNOR bitwise operators
9. LIVE NOT (one's complement) and NEG (two's complement)
10. LIVE Shift left SHL and logical shift right SHR
11. LIVE Arithmetic shift right ASR (keeps the sign bit)
12. LIVE Rotate left ROL and rotate right ROR within the word size
13. LIVE Integer division ÷ (truncates toward zero) and MOD remainder
14. LIVE C-like precedence: × ÷ MOD, then + −, shifts, AND, XOR, OR
15. LIVE Population count POPCNT
16. LIVE Count leading zeros CLZ and trailing zeros CTZ
17. LIVE Parity bit PARITY
18. LIVE Set / clear / flip / test single bits SETB, CLRB, FLIP, TSTB
19. LIVE Byte swap BSWAP (endianness)
20. LIVE Bit reversal RBIT

21. LIVE Sign extension SEXT(x,n)
22. LIVE Lowest and highest set bit LSB / MSB
23. LIVE GCD and ABS on integers
24. LIVE Bit-grid view of the whole word in groups of four
25. LIVE Character view: the result shown as a Unicode character
26. LIVE Character → code table (ASCII)
27. LIVE Ten registers R0–R9 with STO / RCL and a register list
28. LIVE Ans in the current base
29. LIVE Brackets with precedence
30. LIVE History recall ▲ ▼ and cursor editing
31. LIVE Base switch re-displays the last answer without recalculating
32. LIVE Clear error messages for divide by 0 and digits outside the base
33. LIVE Double-zero 00 key for fast entry
34. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
35. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
36. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
37. APP/UI Show / hide key numbers with one button (remembered between sessions)
38. APP/UI Copy the current display text to the clipboard
39. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
40. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen

41. APP/UI **Reference photo viewer and tool-tip name on every key**
42. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
43. PLANNED **Exam mode (locks memory & shows indicator)**
44. PLANNED **Answer history export to text**
45. PLANNED **Speech read-out of results**
46. PLANNED **Colour-blind safe key theme**
47. PLANNED **Custom shortcut keys**

2b · Feature guide

Calculator #26 is a integer algebraic entry in the selected base (HEX/DEC/OCT/BIN), then =. Each card below is one of its 33 working features, with the keys that reach it and, where one exists, a verified example.

F1 Exact integer arithmetic on BigInt values (no rounding at any word size)

Where: #5 WORD Word size 8 → 16 → 32 → 64 bits; #11 ASCII Character → code table; #24 ASR Arithmetic shift right

See it in action: Integer division — *17 ÷ 5* = 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 HEX, DEC, OCT and BIN entry and display modes

Where: #1 HEX Hexadecimal mode; #2 DEC Decimal mode; #3 OCT Octal mode

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Hex digits A–F on dedicated keys (refused in DEC/OCT/BIN)

Where: #1 HEX Hexadecimal mode; #2 DEC Decimal mode; #3 OCT Octal mode

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Simultaneous view of the result in the three other bases

Where: #8 BITS Bit-grid view on/off; #58 = Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Word size 8 / 16 / 32 / 64 bits with wrap-around

Where: #5 WORD Word size 8 → 16 → 32 → 64 bits; #11 ASCII Character → code table; #8 BITS Bit-grid view on/off

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Signed (two's complement) or unsigned interpretation

Where: #6 ±SIGN Signed / unsigned; #11 ASCII Character → code table; #16 NOT One's complement

See it in action: One's complement of 0 (signed) — *NOT 0* = -1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 AND, OR, XOR bitwise operators

Where: #15 XOR Bitwise XOR; #13 AND Bitwise AND; #14 OR Bitwise OR

See it in action: Bitwise AND (decimal) — *12 AND 10* = 8; Bitwise OR — *12 OR 3* = 15

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 NAND, NOR, XNOR bitwise operators

Where: #13 AND Bitwise AND; #14 OR Bitwise OR; #15 XOR Bitwise XOR

See it in action: Bitwise AND (decimal) — *12 AND 10* = 8; Bitwise OR — *12 OR 3* = 15

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 NOT (one's complement) and NEG (two's complement)

Where: #16 NOT One's complement; #53 NEG Negative sign (–) — makes the next number negative. Different from the subtraction key

See it in action: One's complement of 0 (signed) — *NOT 0* = -1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Shift left SHL and logical shift right SHR

Where: #18 SHR Logical shift right; #17 SHL Shift left; #24 ASR Arithmetic shift right

See it in action: Shift left — *1 SHL 10* = 1024

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Arithmetic shift right ASR (keeps the sign bit)

Where: #24 ASR Arithmetic shift right; #18 SHR Logical shift right; #6 ±SIGN Signed / unsigned

See it in action: Shift left — *1 SHL 10* = 1024

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Rotate left ROL and rotate right ROR within the word size

Where: #17 SHL Shift left; #18 SHR Logical shift right; #5 WORD Word size 8 → 16 → 32 → 64 bits

See it in action: Shift left — *1 SHL 10* = 1024

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Integer division ÷ (truncates toward zero) and MOD remainder

Where: #34 MOD Remainder (modulo); #40 ÷ Division ÷

See it in action: Remainder — *100 MOD 7* = 2; Integer division — *17 ÷ 5* = 3

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 C-like precedence: × ÷ MOD, then + −, shifts, AND, XOR, OR

Where: #15 XOR Bitwise XOR; #34 MOD Remainder (modulo)

See it in action: Bitwise XOR — *15 XOR 5* = 10; Remainder — *100 MOD 7* = 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Population count POPCNT

Where: #19 POPC Count 1-bits; #20 CLZ Count leading zeros

See it in action: Population count — *POPCNT(255)* = 8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Count leading zeros CLZ and trailing zeros CTZ

Where: #20 CLZ Count leading zeros; #56 00 Two zeros; #19 POPC Count 1-bits

See it in action: Population count — *POPCNT(255)*= 8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Parity bit PARITY

Where: #19 POPC Count 1-bits; #8 BITS Bit-grid view on/off; #9 FUNC Bit-function catalogue

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Set / clear / flip / test single bits SETB, CLRB, FLIP, TSTB

Where: #21 SETB Set bit n; #22 FLIP Flip bit n; #5 WORD Word size 8 → 16 → 32 → 64 bits

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Byte swap BSWAP (endianness)

Where: #23 BSWP Byte swap

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Bit reversal RBIT

Where: #8 BITS Bit-grid view on/off; #9 FUNC Bit-function catalogue; #21 SETB Set bit n

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Sign extension SEXT(x,n)

Where: #6 ±SIGN Signed / unsigned; #53 NEG Negative sign (–) — makes the next number negative. Different from the subtraction key

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Lowest and highest set bit LSB / MSB

Where: #21 SETB Set bit n; #8 BITS Bit-grid view on/off; #9 FUNC Bit-function catalogue

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 GCD and ABS on integers

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Bit-grid view of the whole word in groups of four

Where: #8 BITS Bit-grid view on/off; #5 WORD Word size 8 → 16 → 32 → 64 bits; #9 FUNC Bit-function catalogue

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Character view: the result shown as a Unicode character

Where: #8 BITS Bit-grid view on/off; #11 ASCII Character → code table; #30 DEL DEL — delete the character at the cursor

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Character → code table (ASCII)

Where: #11 ASCII Character → code table; #8 BITS Bit-grid view on/off; #30 DEL DEL — delete the character at the cursor

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Ten registers R0–R9 with STO / RCL and a register list

Where: #10 REGS Show registers R0–R9; #35 , Comma / argument separator (between function arguments, list items or coordinates); #41 STO STO — store the value into a memory / variable (press the target next)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Ans in the current base

Where: #54 Ans Ans — recall the last answer into the current expression; #36 AC AC — clear all of the current calculation

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Brackets with precedence

Where: #29) Closing bracket). Closing brackets at the end of a line may be omitted

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 History recall ▲ ▼ and cursor editing

Where: #54 Ans Ans — recall the last answer into the current expression; #30 DEL DEL — delete the character at the cursor; #42 RCL RCL — recall a memory / variable

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Base switch re-displays the last answer without recalculating

Where: #54 Ans Ans — recall the last answer into the current expression

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F32 Clear error messages for divide by 0 and digits outside the base

Where: #10 REGS Show registers R0–R9; #21 SETB Set bit n; #36 AC AC — clear all of the current calculation

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F33 Double-zero 00 key for fast entry

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #26 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	HEX	·	·	·	fn
2	DEC	·	·	·	fn
3	OCT	·	·	·	fn
4	BIN	·	·	·	fn
5	WORD	·	·	·	fn
6	±SIGN	·	·	·	fn
7	SHIFT	·	·	·	mod
8	BITS	CHR	·	·	fn
9	FUNC	·	·	·	fn
10	REGS	CLR R	·	·	fn
11	ASCII	SETUP	·	·	fn
12	ON	·	·	·	clr
13	AND	NAND	·	·	fn
14	OR	NOR	·	·	fn
15	XOR	XNOR	·	·	fn
16	NOT	·	·	·	fn
17	SHL	ROL	·	·	fn
18	SHR	ROR	·	·	fn
19	POPC	PAR	·	·	fn
20	CLZ	CTZ	·	·	fn
21	SETB	CLRB	·	·	fn
22	FLIP	TSTB	·	·	fn
23	BSWP	RBIT	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	ASR	·	·	·	fn
25	D	·	·	·	num
26	E	·	·	·	num
27	F	·	·	·	num
28	(	·	·	·	op
29	)	·	·	·	op
30	DEL	·	·	·	clr
31	A	·	·	·	num
32	B	·	·	·	num
33	C	·	·	·	num
34	MOD	·	·	·	fn
35	,	·	·	·	op
36	AC	·	·	·	clr
37	7	·	·	·	num
38	8	·	·	·	num
39	9	·	·	·	num
40	÷	·	·	·	op
41	STO	·	·	·	mem
42	RCL	·	·	·	mem
43	4	·	·	·	num
44	5	·	·	·	num
45	6	·	·	·	num
46	×	·	·	·	op
47	◀	·	·	·	mod
48	▶	·	·	·	mod
49	1	·	·	·	num
50	2	·	·	·	num
51	3	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	−	·	·	·	op
53	NEG	·	·	·	num
54	Ans	▲	·	·	mem
55	0	·	·	·	num
56	00	·	·	·	fn
57	+	·	·	·	op
58	=	·	·	·	op

4 · Key-by-key reference

Every key on calculator #26, in number order, with every function it carries in this app.

#1 HEX Functions

Primary: Hexadecimal mode

#2 DEC Functions

Primary: Decimal mode

#3 OCT Functions

Primary: Octal mode

#4 BIN Functions

Primary: Binary mode

#5 WORD Functions

Primary: Word size 8 → 16 → 32 → 64 bits

#6 ±SIGN Functions

Primary: Signed / unsigned

#7 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#8 BITS `Functions`

Primary: Bit-grid view on/off

shifted (2nd / f / SHIFT layer) (press #7 first): CHR — Character view on/off

#9 FUNC `Functions`

Primary: Bit-function catalogue

#10 REGS `Functions`

Primary: Show registers R0–R9

shifted (2nd / f / SHIFT layer) (press #7 first): CLR R — Clear all registers

#11 ASCII `Functions`

Primary: Character → code table

shifted (2nd / f / SHIFT layer) (press #7 first): SETUP — Word size / signed menu

#12 ON `Clear, delete & power`

Primary: ON — power on

#13 AND Functions

Primary: Bitwise AND

shifted (2nd / f / SHIFT layer) (press #7 first): NAND — Bitwise NAND

#14 OR Functions

Primary: Bitwise OR

shifted (2nd / f / SHIFT layer) (press #7 first): NOR — Bitwise NOR

#15 XOR Functions

Primary: Bitwise XOR

shifted (2nd / f / SHIFT layer) (press #7 first): XNOR — Bitwise XNOR

#16 NOT Functions

Primary: One's complement

#17 SHL Functions

Primary: Shift left

shifted (2nd / f / SHIFT layer) (press #7 first): ROL — Rotate left

#18 SHR Functions

Primary: Logical shift right

shifted (2nd / f / SHIFT layer) (press #7 first): ROR — Rotate right

#19 POPC Functions

Primary: Count 1-bits

shifted (2nd / f / SHIFT layer) (press #7 first): PAR — Parity

#20 CLZ Functions

Primary: Count leading zeros

shifted (2nd / f / SHIFT layer) (press #7 first): CTZ — Count trailing zeros

#21 SETB Functions

Primary: Set bit n

shifted (2nd / f / SHIFT layer) (press #7 first): CLRB — Clear bit n

#22 FLIP Functions

Primary: Flip bit n

shifted (2nd / f / SHIFT layer) (press #7 first): TSTB — Test bit n

#23 BSWP Functions

Primary: Byte swap

shifted (2nd / f / SHIFT layer) (press #7 first): RBIT — Reverse bits

#24 ASR Functions

Primary: Arithmetic shift right

#25 D `Digits & number entry`

Primary: Hex digit D

#26 E `Digits & number entry`

Primary: Hex digit E

#27 F `Digits & number entry`

Primary: Hex digit F

#28 (`Operators & execute`

Primary: Opening bracket (

Keyboard: (

#29) `Operators & execute`

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#30 DEL `Clear, delete & power`

Primary: DEL — delete the character at the cursor

Keyboard: `Backspace` `Delete`

#31 A `Digits & number entry`

Primary: Hex digit A

#32 B `Digits & number entry`

Primary: Hex digit B

#33 C `Digits & number entry`

Primary: Hex digit C

#34 MOD `Functions`

Primary: Remainder (modulo)

#35 , `Operators & execute`

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: `,`

#36 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: `Escape`

#37 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#38 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#39 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#40 ÷ Operators & execute

Primary: Division ÷

Keyboard: /

#41 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

#42 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#43 **4** `Digits & number entry`

Primary: Digit 4

Keyboard: 4

#44 **5** `Digits & number entry`

Primary: Digit 5

Keyboard: 5

#45 **6** `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#46 **×** `Operators & execute`

Primary: Multiplication ×

Keyboard: * x

#47 **◀** `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

Keyboard: `ArrowLeft`

#48 **▶** `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

Keyboard: `ArrowRight`

#49 **1** `Digits & number entry`

Primary: Digit 1

Keyboard: 1

#50 **2** `Digits & number entry`

Primary: Digit 2

Keyboard: 2

#51 **3** `Digits & number entry`

Primary: Digit 3

Keyboard: 3

#52 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: -

#53 **NEG** `Digits & number entry`

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

#54 Ans `Memory, variables & registers`

Primary: Ans — recall the last answer into the current expression

shifted (2nd / f / SHIFT layer) (press #7 first): ▲ — Cursor ▲ — up / previous history line / previous item

#55 0 `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#56 00 `Functions`

Primary: Two zeros

#57 + `Operators & execute`

Primary: Addition +

Keyboard: `+`

#58 = `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

5 · Worked examples

8 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Bitwise AND (decimal)

Calculate **12 AND 10**. Press **Reset** first.

#49 1 → #50 2 → #13 AND → #49 1 → #55 0 → #58 =

Display shows 8 — expected 8. ✔ verified on the app.

Example 5.2 — Bitwise OR

Calculate **12 OR 3**. Press **Reset** first.

#49 1 → #50 2 → #14 OR → #51 3 → #58 =

Display shows 15 — expected 15. ✔ verified on the app.

Example 5.3 — Bitwise XOR

Calculate **15 XOR 5**. Press **Reset** first.

#49 1 → #44 5 → #15 XOR → #44 5 → #58 =

Display shows 10 — expected 10. ✔ verified on the app.

Example 5.4 — Shift left

Calculate **1 SHL 10**. Press **Reset** first.

`#49` 1 → `#17` SHL → `#49` 1 → `#55` 0 → `#58` =

Display shows `1024` — expected 1024. ✔ verified on the app.

Example 5.5 — Remainder

Calculate **100 MOD 7**. Press **Reset** first.

`#49` 1 → `#55` 0 → `#55` 0 → `#34` MOD → `#37` 7 → `#58` =

Display shows `2` — expected 2. ✔ verified on the app.

Example 5.6 — One's complement of 0 (signed)

Calculate **NOT 0**. Press **Reset** first.

`#16` NOT → `#55` 0 → `#58` =

Display shows `-1` — expected -1. ✔ verified on the app.

Example 5.7 — Integer division

Calculate **17 ÷ 5**. Press **Reset** first.

`#49` 1 → `#37` 7 → `#40` ÷ → `#44` 5 → `#58` =

Display shows `3` — expected 3. ✔ verified on the app.

Example 5.8 — Population count

Calculate **POPCNT(255)**. Press **Reset** first.



Display shows 8 — expected 8. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

14 extra functions sit on shifted layers. Press the layer key once, then the key: for example #7 then #8 → CHR; #7 then #10 → CLR R; #7 then #11 → SETUP; #7 then #13 → NAND.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #58.

Hack 13: Type faster on a computer

22 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
`(`	#28	(	Opening bracket (
`)`	#29	)	Closing bracket). Closing brackets at the end of a line may be omitted
`Backspace` `Delete`	#30	DEL	DEL — delete the character at the cursor
`,`	#35	,	Comma / argument separator (between function arguments, list items or coordinates)
`Escape`	#36	AC	AC — clear all of the current calculation
`7`	#37	7	Digit 7
`8`	#38	8	Digit 8
`9`	#39	9	Digit 9
`/`	#40	÷	Division ÷
`4`	#43	4	Digit 4
`5`	#44	5	Digit 5
`6`	#45	6	Digit 6
`*` `x`	#46	×	Multiplication ×
`ArrowLeft`	#47	◀	Cursor ◀ — move left in the line
`ArrowRight`	#48	▶	Cursor ▶ — move right in the line
`1`	#49	1	Digit 1
`2`	#50	2	Digit 2
`3`	#51	3	Digit 3
`-`	#52	−	Subtraction −
`0`	#55	0	Digit 0

+	#57	+	Addition +
Enter =	#58	=	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 9: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (8)

Work each one on calculator #26, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 12 AND 10 = _____
2. 12 OR 3 = _____
3. 15 XOR 5 = _____
4. 1 SHL 10 = _____
5. 100 MOD 7 = _____
6. NOT 0 = _____
7. 17 ÷ 5 = _____
8. POPCNT(255) = _____

Set B — Key-finding drills (62)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “HEX” (Hexadecimal mode)? _____
2. Which key is “DEC” (Decimal mode)? _____
3. Which key is “OCT” (Octal mode)? _____
4. Which key is “BIN” (Binary mode)? _____
5. Which key is “WORD” (Word size 8 → 16 → 32 → 64 bits)? _____
6. Which key is “±SIGN” (Signed / unsigned)? _____

7. Which key is “SHIFT” (SHIFT / 2nd)? _____

8. Which key is “BITS” (Bit-grid view on/off)? _____

9. Which key is “FUNC” (Bit-function catalogue)? _____

10. Which key is “REGS” (Show registers R0–R9)? _____

11. Which key is “ASCII” (Character → code table)? _____

12. Which key is “ON” (ON)? _____

13. Which key is “AND” (Bitwise AND)? _____

14. Which key is “OR” (Bitwise OR)? _____

15. Which key is “XOR” (Bitwise XOR)? _____

16. Which key is “NOT” (One’s complement)? _____

17. Which key is “SHL” (Shift left)? _____

18. Which key is “SHR” (Logical shift right)? _____

19. Which key is “POPC” (Count 1-bits)? _____

20. Which key is “CLZ” (Count leading zeros)? _____

21. Which key is “SETB” (Set bit n)? _____

22. Which key is “FLIP” (Flip bit n)? _____

23. Which key is “BSWP” (Byte swap)? _____

24. Which key is “ASR” (Arithmetic shift right)? _____

25. Which key is “D” (Hex digit D)? _____

26. Which key is “E” (Hex digit E)? _____

27. Which key is “F” (Hex digit F)? _____

28. Which key is “(” (Opening bracket)? _____

29. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

30. Which key is “DEL” (DEL)? _____

31. Which key is “A” (Hex digit A)? _____

32. Which key is “B” (Hex digit B)? _____

33. Which key is “C” (Hex digit C)? _____

34. Which key is “MOD” (Remainder)? _____

35. Which key is “,” (Comma / argument separator)? _____

36. Which key is “AC” (AC)? _____

37. Which key is “÷” (Division ÷)? _____

38. Which key is “STO” (STO)? _____

39. Which key is “RCL” (RCL)? _____

40. Which key is “×” (Multiplication ×)? _____

41. Which key is “◀” (Cursor ◀)? _____

42. Which key is “▶” (Cursor ▶)? _____

43. Which key is “−” (Subtraction −)? _____

44. Which key is “NEG” (Negative sign)? _____

45. Which key is “Ans” (Ans)? _____

46. Which key is “00” (Two zeros)? _____

47. Which key is “+” (Addition +)? _____

48. Which key is “=” (Execute / =)? _____

49. How do you reach “CHR”? _____

50. How do you reach “CLR R”? _____

51. How do you reach “SETUP”? _____

52. How do you reach “NAND”? _____

53. How do you reach “NOR”? _____

54. How do you reach “XNOR”? _____

55. How do you reach “ROL”? _____

56. How do you reach “ROR”? _____

57. How do you reach “PAR”? _____

58. How do you reach “CTZ”? _____

59. How do you reach “CLRB”? _____

60. How do you reach “TSTB”? _____

61. How do you reach “RBIT”? _____

62. How do you reach “ ▲ ”? _____

Appendix A — Drill answers

Set A

1. **8** (display `8`)

   `#49` 1 → `#50` 2 → `#13` AND → `#49` 1 → `#55` 0 → `#58` =

2. **15** (display `15`)

   `#49` 1 → `#50` 2 → `#14` OR → `#51` 3 → `#58` =

3. **10** (display `10`)

   `#49` 1 → `#44` 5 → `#15` XOR → `#44` 5 → `#58` =

4. **1024** (display `1024`)

   `#49` 1 → `#17` SHL → `#49` 1 → `#55` 0 → `#58` =

5. **2** (display `2`)

   `#49` 1 → `#55` 0 → `#55` 0 → `#34` MOD → `#37` 7 → `#58` =

6. **-1** (display `-1`)

   `#16` NOT → `#55` 0 → `#58` =

7. **3** (display `3`)

   `#49` 1 → `#37` 7 → `#40` ÷ → `#44` 5 → `#58` =

8. **8** (display `8`)

   `#19` POPC → `#50` 2 → `#44` 5 → `#44` 5 → `#29`) → `#58` =

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #40
38. #41
39. #42
40. #46
41. #47
42. #48
43. #52
44. #53
45. #54
46. #56
47. #57
48. #58
49. #7 SHIFT then #8 BITS
50. #7 SHIFT then #10 REGS
51. #7 SHIFT then #11 ASCII
52. #7 SHIFT then #13 AND
53. #7 SHIFT then #14 OR
54. #7 SHIFT then #15 XOR
55. #7 SHIFT then #17 SHL
56. #7 SHIFT then #18 SHR
57. #7 SHIFT then #19 POPC
58. #7 SHIFT then #20 CLZ
59. #7 SHIFT then #21 SETB
60. #7 SHIFT then #22 FLIP
61. #7 SHIFT then #23 BSWP
62. #7 SHIFT then #54 Ans

Appendix B — Function finder (A–Z)

Function	Label	Keys
AC — clear all of the current calculation	AC	#36
Addition +	+	#57
Ans — recall the last answer into the current expression	Ans	#54
Arithmetic shift right	ASR	#24
Binary mode	BIN	#4
Bit-function catalogue	FUNC	#9
Bit-grid view on/off	BITS	#8
Bitwise AND	AND	#13
Bitwise NAND	NAND	#7 → #13
Bitwise NOR	NOR	#7 → #14
Bitwise OR	OR	#14
Bitwise XNOR	XNOR	#7 → #15
Bitwise XOR	XOR	#15
Byte swap	BSWP	#23
Character → code table	ASCII	#11
Character view on/off	CHR	#7 → #8
Clear all registers	CLR R	#7 → #10
Clear bit n	CLRB	#7 → #21
Closing bracket). Closing brackets at the end of a line may be omitted	)	#29
Comma / argument separator (between function arguments, list items or coordinates)	,	#35
Count 1-bits	POPC	#19

Count leading zeros	CLZ	#20
Count trailing zeros	CTZ	#7 → #20
Cursor ▲ — up / previous history line / previous item	▲	#7 → #54
Cursor ▶ — move right in the line	▶	#48
Cursor ◀ — move left in the line	◀	#47
Decimal mode	DEC	#2
DEL — delete the character at the cursor	DEL	#30
Digit 0	0	#55
Digit 1	1	#49
Digit 2	2	#50
Digit 3	3	#51
Digit 4	4	#43
Digit 5	5	#44
Digit 6	6	#45
Digit 7	7	#37
Digit 8	8	#38
Digit 9	9	#39
Division ÷	÷	#40
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	=	#58
Flip bit n	FLIP	#22
Hex digit A	A	#31
Hex digit B	B	#32
Hex digit C	C	#33
Hex digit D	D	#25
Hex digit E	E	#26
Hex digit F	F	#27
Hexadecimal mode	HEX	#1

Logical shift right	SHR	#18
Multiplication ×	×	#46
Negative sign (−) — makes the next number negative. Different from the subtraction key	NEG	#53
Octal mode	OCT	#3
ON — power on	ON	#12
One's complement	NOT	#16
Opening bracket (	(	#28
Parity	PAR	#7 → #19
RCL — recall a memory / variable	RCL	#42
Remainder (modulo)	MOD	#34
Reverse bits	RBIT	#7 → #23
Rotate left	ROL	#7 → #17
Rotate right	ROR	#7 → #18
Set bit n	SETB	#21
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#7
Shift left	SHL	#17
Show registers R0–R9	REGS	#10
Signed / unsigned	±SIGN	#6
STO — store the value into a memory / variable (press the target next)	STO	#41
Subtraction −	−	#52
Test bit n	TSTB	#7 → #22
Two zeros	00	#56
Word size / signed menu	SETUP	#7 → #11
Word size 8 → 16 → 32 → 64 bits	WORD	#5

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Programmer calculator working on exact integers (JavaScript BigInt) with a selectable word size. The evaluator, bit functions and display are original U-45 code.

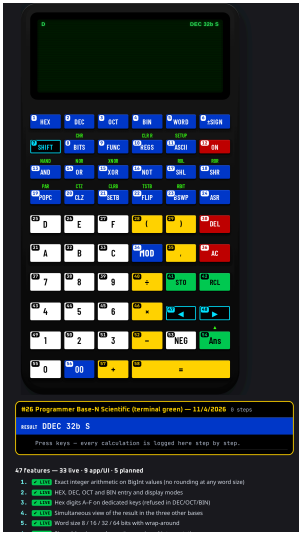
DEC mode shows signed values when SIGNED is on; HEX/OCT/BIN always show the raw bit pattern of the word.

Planned (not yet working)

- **Exam mode (locks memory & shows indicator)**
- **Answer history export to text**
- **Speech read-out of results**
- **Colour-blind safe key theme**
- **Custom shortcut keys**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 HEX	#2 DEC	#3 OCT
#4 BIN	#5 WORD	#6 ±SIGN
#7 SHIFT	#8 BITS ⇧ CHR	#9 FUNC
#10 REGS ⇧ CLR R	#11 ASCII ⇧ SETUP	#12 ON
#13 AND ⇧ NAND	#14 OR ⇧ NOR	#15 XOR ⇧ XNOR
#16 NOT	#17 SHL ⇧ ROL	#18 SHR ⇧ ROR
#19 POPC ⇧ PAR	#20 CLZ ⇧ CTZ	#21 SETB ⇧ CLRB
#22 FLIP ⇧ TSTB	#23 BSWP ⇧ RBIT	#24 ASR
#25 D	#26 E	#27 F
#28 (	#29)	#30 DEL
#31 A	#32 B	#33 C
#34 MOD	#35 ,	#36 AC
#37 7	#38 8	#39 9
#40 ÷	#41 STO	#42 RCL
#43 4	#44 5	#45 6
#46 ×	#47 ◀	#48 ▶
#49 1	#50 2	#51 3

#52 –	**#53** NEG	**#54** Ans ⇧ ▲
#55 0	**#56** 00	**#57** +
#58 =		

U-45 CALCULATORS · Calculator #26 · cut out and keep